

Exploration of Visuo-haptic Interactions to Support Learning Leopold's Maneuvers Process in Virtual Reality

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ABSTRACT

We present Leopold's Maneuvers VR, a haptic-enabled virtual reality simulation in which a user determines the size, position, and weight of a fetus within a virtual patient by palpating the patient's abdomen. Users of the application receive corresponding haptic cues (force and vibration) as they touch the fetus through the patient's torso. The physical sensations generated by the SenseGlove, paired with the immersive visuals within the virtual environment, support to create a palpation experience similar to the real Leopold's Maneuvers activity. This application addresses a need in nursing education for an immersive virtual reality experience that integrates direct hand manipulation in the learning of assessment skills.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); Interaction paradigms; Virtual reality.

KEYWORDS

Virtual Reality, Visuo-haptic feedback, Haptic Interaction, Leopold's Maneuvers, Nursing Education

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1 INTRODUCTION

Immersive technologies have been utilized in education and other disciplines to support students' learning. Virtual Reality (VR) allows users to directly interact with virtual representations of real objects, visualize abstract concepts, articulate the learners' understanding, and enhance competency and critical thinking skills [Kilmon et al., 2010; Kaddoura, 2010]. Recently, VR has been heavily utilized in

health-related professions. Nursing education adopts VR technology to enable students to learn essential skills, improve decision making and critical thinking in a simulated environment. VR also provides a variety of patient scenarios that allow students to practice their skills repetitively in a safe environment [Pottle, 2019]. However, most VR applications focus on visual and sound stimuli, which limit multisensory and embodied experiences for students. Previous research has demonstrated that incorporating haptic interactions in VR enhances user's experience, immersion, realism, and task performance and allows a high degree of presence [Panait et al., 2009].

For the XR demo at SIGGRAPH Asia, we will present a haptic-enabled VR simulation project, called Leopold's Maneuvers VR that allows nursing students to practice the Leopold's Maneuvers process. The project aims to provide students with a dynamic and immersive learning experience by enabling users to touch and see the virtual patient's abdomen. The application supports a user's visual and muscle memories in the immersive learning environment, providing multisensory feedback such as: force feedback, tactile feedback, and visual feedback.

This application was implemented using the HTC VIVE head mounted display (HMD) and the SenseGlove Development Kit [SenseGlove, 2021] in a Unity 3D environment. We created our virtual environment using custom designed assets from Autodesk Maya.

2 CHALLENGES IN LEOPOLD'S MANEUVERS

Leopold's Maneuvers is a systematic method of observation and palpation to determine fetal position, presentation, lie, and attitude, which helps predict the course of labor (Figure 1). A healthcare provider performs the maneuvers by using their fingers and entire palm to palpate on the patient's abdomen. This is a difficult task for nurses and providers because they can only assess the fetus's size, location, and direction by touching without seeing a fetus directly. When performing the maneuvers, it is critical to ensure patient safety and minimize the risk of harm. Generally, this training has been done with a plastic abdominal palpation model or a standardized patient (SP). The palpation model provides a semi-realistic abdominal shape and a fetus model. However, students do not receive immediate and interactive feedback of the fetus movement. Incorporating SPs in the assessment is a more effective way

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Figure 1: A nurse conducting Leopold's Maneuvers



Figure 2: Two different visual modes

to obtain hands-on experience, but they are difficult to find, time-consuming, and costly. Due to the pandemic, it has become more difficult to organize SPs for training. A multisensory experience within a VR environment can help students to gain hands-on experience and recreate this experience in a more realistic and interactive way.

3 DESIGN OF LEOPOLD'S MANEUVERS VR

Leopold's Maneuvers VR was designed to provide nursing students with an interactive and immersive learning experience in the Leopold's Maneuvers process. This application provides unique visual feedback that would normally be inaccessible in a real-life situation as well as haptic feedback that would not be accessed in typical learning environments. In the X-ray mode of the application, users can see the head and body of the fetus as they touch the mother's abdomen and allows them to associate their sense of touch and spatial understanding with anatomical learning information. In traditional learning material (like 2D diagrams and videos), students cannot easily associate the information on paper to what they touch in person.

3.1 Visual Interaction Design

In the application, a user starts in a virtual examination room with a pregnant woman on a patient bed. The room was designed to



Figure 3: The haptic gloves

replicate a typical examination room to enhance the user's immersion in the VR experience. After a brief video introduction of the application, the user washes their hands and starts the maneuver process. While performing the maneuver, the user can touch a virtual patient's abdomen, view visual changes on the abdomen area, and feel the fetal presence via the haptic gloves. We provide two different visual modes (transparent vs. normal, Figure 2) for displaying the abdomen and fetus, allowing instructors to test how each visualization mode impacts students' learning and immersion. The transparent mode provides unique visual feedback that is normally inaccessible in real life. The user can investigate the fetus's movement more carefully while they are conducting the maneuvers.

3.2 Haptic Interaction Design

Leopold's Maneuvers VR utilized a pair of haptic gloves, specifically Development Kit from SenseGlove, to add an extra sensory dimension to the virtual environment (Figure 3). With these gloves, the user can physically detect different levels of resistance (soft, medium, and hard) through their fingers depending on the material quality (hardness) of the virtual body. They also feel a gentle vibration on the fingertips when the virtual hand touches the surface of the virtual objects, further reinforcing the tactile experience. The gloves have an exoskeleton on each finger that provides force feedback by pulling on the fingers, and vibro-tactile feedback through vibration motors on each fingertip. In terms of the force feedback of the SenseGlove, the maximum force applied to each fingertip is 40N (4 kg). In the Unity environment, we assigned a different force-feedback intensity (0-100%) to each material to create a distinction between hard and soft materials. 50% of force intensity was applied to the belly surface, while 90% of force was applied to the fetus, making it feel "harder". The vibration provides an additional sensation to amplify the experience. We tested vibration intensity values for each different surface material and found that 60% vibration intensity was appropriate for the belly and 95% for

the fetus. By pairing these two feedback systems together, the users can feel and touch the virtual patient's abdomen in a more natural and immersive way. The use of haptic gloves in a VR simulation would be beneficial in a variety of circumstances, especially when performing actions like Leopold's Maneuvers, as touching and palpating a patient's abdomen is an integral part of the procedure. The haptic gloves help the user feel the shape of the fetus, which is supported by the visual deformation of the belly's surface when it is pushed on by the user. In our system, the user can safely assess the patient's condition and make decisions on how much pressure they can put on the fetus without harming either the mother or fetus. From the preliminary testing of the system with nursing faculty and students, we learned that the haptic sensations support a similar bodily experience to the real palpation activity in Leopold's Maneuvers. The audio narration by a nursing faculty member in the application helped the users follow the steps of the procedure more easily. There were also a few limitations reported by the test users: the mechanical noise from the gloves broke the participant's immersion, and the weight of the gloves and the trackers caused the users' arms/hands to be tired. The physical fatigue from the weight of the equipment cannot be solved right away but we will

integrate a newer version of the gloves which are lighter and more comfortable.

4 CONCLUSION

We presented a VR application that enables students to experience the Leopold's Maneuvers assessment process with realistic visual and haptic feedback in a highly interactive and immersive environment. Preliminary testing with nursing faculty and students show potential for this application to provide a high-quality learning environment for students in women's health education.

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